Module 2 - Dashboard Creation

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ALY6110: Data Management and Big Data

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April 19th 2026

Introduction

In the UK, road safety is still a major public health concern, and although cycling is becoming more and more popular as a form of transportation, there are risks involved. An exploratory data analysis of bicycle accidents in Great Britain over a four-decade period, from 1979 to 2018, is presented in this research. Over 827,000 rows of accident-level data with features encompassing temporal patterns, geographic context, fatality demographics, and accident severity make up the dataset, which was first assembled from UK Department for Transport records and was sourced from Kaggle.

Using descriptive statistics and interactive visualizations, this investigation seeks to identify significant trends, patterns, and risk factors related to bicycle accidents. To display these results in an approachable, interactive way that enables users to filter, drill down, and investigate the data independently, a Power BI dashboard was created.

Dataset Overview

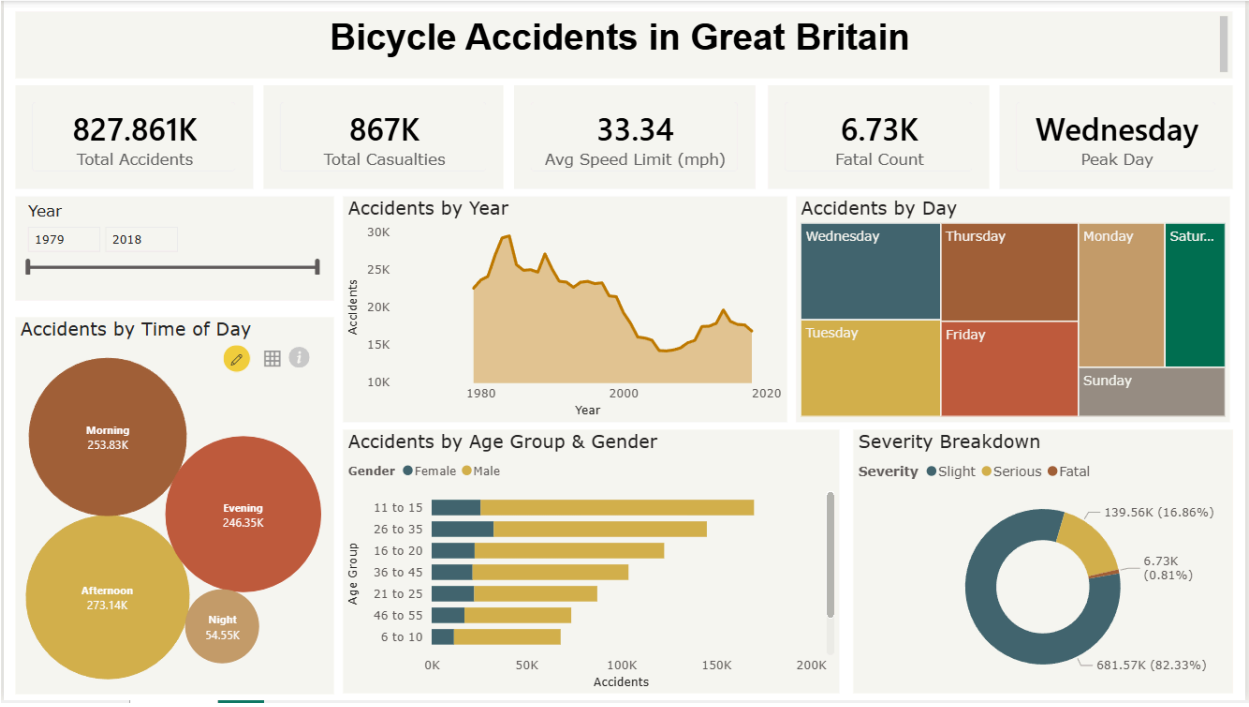
The dataset consists of 827,861 recorded bicycle accidents across Great Britain. Each record includes details such as the year and day of the accident, time of day, casualty age group, gender, speed limit of the road, and severity classification (slight, serious, or fatal). A year slicer covering 1979 to 2018 enables temporal filtering across the entire dashboard.

Key Metrics:

Metric	Value
Total Accidents	827,861
Total Casualties	867,000
Average Speed Limit	33.34 mph
Fatal Accidents	6,730
Most Common Day	Wednesday

Dashboard & Visualizations

The Power BI dashboard consolidates all key findings into a single interactive view. It includes KPI cards for headline metrics, a year slicer for temporal filtering, and six visualizations covering trends over time, day-of-week distribution, time-of-day patterns, demographic breakdowns, and severity composition.



1. **Accidents by Year**

The line chart shows a clear long-term decline in bicycle accidents. Accidents peaked at approximately 30,000 per year in the early 1980s and fell to roughly 15,000–18,000 by the 2010s. This downward trend likely reflects improvements in road infrastructure, traffic calming measures, and cycling safety campaigns introduced over the decades. However, there is a notable uptick around 2010-2012, possibly corresponding to cycling popularity surges tied to events like the London Olympics.

2. **Day-of-Week Distribution**

The treemap visualization shows that accidents are distributed relatively evenly across weekdays, with Wednesday emerging as the day with the highest count. Weekends, particularly Sunday see noticeably fewer accidents, which aligns with reduced commuter traffic. This pattern suggests that weekday commuting is a primary context for bicycle accidents, and targeted interventions during peak commute hours on weekdays could have the most impact.

3. **Time-of-Day Patterns**

The bubble chart breaks accidents into four time periods: Morning (253.83K), Afternoon (273.14K), Evening (246.35K), and Night (54.55K). The afternoon period accounts for the highest share of accidents, followed closely by morning and evening. Night-time accidents are significantly lower in volume but may carry disproportionately higher severity due to reduced visibility. Together, the morning and afternoon periods typical commute and school-run hours account for over 63% of all recorded accidents.

4. **Demographics (Age Group & Gender)**

Accidents are broken down by gender and age group in the horizontal bar chart. The age group of 11 to 15 has the highest number of accidents, which is alarming since it shows how vulnerable school-age youngsters are on the roadways. The 26–35 and 16–20 age groups follow closely, representing young adults and teenagers. Males are overrepresented across every age group, consistent with broader cycling participation data showing higher ridership among men. The 6–10, 46–55 and higher age groups have the lowest counts, reflecting lower cycling volumes at those life stages.

5. **Severity Breakdown**

The donut chart shows that the vast majority of accidents - 681,570 or 82.33% are classified as slight. Serious injuries account for 139,560 cases (16.86%), while fatal accidents make up 6,730 (0.81%). Although the fatal proportion is small in percentage terms, the absolute number of 6,730 deaths over four decades underscores the ongoing lethality of road conditions for cyclists. Prescriptive recommendations should focus on reducing the serious and fatal categories through infrastructure improvements at high-risk junctions and enforcement of speed limits in cycling corridors.

Conclusion

This analysis of over 827,000 bicycle accident records spanning 1979–2018 reveals a broadly positive long-term trend: accident volumes have roughly halved over four decades. However, significant risks persist, particularly for young cyclists, male riders, and during weekday commuting hours. The Power BI dashboard provides an interactive tool for stakeholders to explore these patterns further, filter by year, and identify areas for targeted intervention. Combining descriptive insights with prescriptive recommendations, this report aims to support evidence-based decision-making for improving cycling safety across Great Britain.

References

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